MIL-C-22546B
26 March 1984
SUPERSEDING
MIL-C-22546A
25 February 1972

MILITARY SPECIFICATION

COCKS, AIR, CUT-OUT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two-way, two-port, compressed-air cut-out cocks having bronze bodies and lever-type operating handles.

1.2 Classification. The air cocks shall be the following types, classes, and sizes, as specified (see 6.2):

Type I - For compressed air supply systems.

Class 150 - 150 pounds per square inch gage (psig) maximum working pressure.
Class 300 - 300 psig maximum working pressure.
Size -
1/2 inch
3/4 inch

Type II - For automotive air brake systems.

Beneficial comments (recommendations, additions, deletions) and any pertienent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4820

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-V-3	- Valves, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-S-81733	- Sealing and Coating Compound Corrosion Inhibitive.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B61 - Steam or Valve Bronze Castings.
ASTM B62 - Composition Bronze or Ounce Metal Castings.
ASTM B584 - Copper Alloy Sand Castings for General Applications.
ASTM D3951 - Standard Practice for Commercial Packaging.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-25 - Standard Marking System for Valves, Fittings, Flanges and Unions.

(Application for copies should be addressed to the Manufacturers Standardization Society of the Valve and Fittings Industry, Incorporated (MSS), 5203 Leesburg Pike, Suite 502, Falls Church, VA 22041.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The air cocks shall consist essentially of a body, a ported plug or ball, and an operating handle. The cocks shall revert from a fully-opened to a fully-closed position and vice versa by rotating the operating handle 90 degrees. Stop lugs for positive positioning of the plug or ball in the open and closed positions shall be provided.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material for components of the air cocks shall be as specified herein. When material for a part is not definitely specified, the material shall be of the type normally used by the contractor in cocks having the same body material and pressure rating as specified herein (see 6.4).

3.3.1 Material deterioration and control. The air cocks shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

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3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.4 Design and performance. The air cocks shall be designed for compressed air service at the applicable nonshock working pressure specified herein. The air cocks shall withstand a shell test pressure equal to 150 percent of the rated working pressure. When the air cocks are tested as specified in 4.5.2.2, internal leakage past the plug or ball in the closed position at the maximum rated working pressure shall not exceed 10.5 cubic inches of free air per minute (equivalent to a one-inch soap bubble in 3 seconds). The torque required to move the valve handle from a fully-opened to a fully-closed position shall not exceed 150 inch-pounds. The area of the passage through the air cock at any point, including the port in the plug or ball, shall be not less than the area of a circle having a diameter equal to the nominal pipe size designation of the air cock.

3.5 Detail of components.

3.5.1 Body. The body shall be copper alloy conforming to ASTM B61, ASTM B62, or ASTM B584, UNS No. 84800. The body shall be furnished with female end connections threaded with American Standard taper pipe threads. Paralled-faced wrenching surfaces shall be provided at each end connection. The body shall be designed to permit access for removal and replacement of the closure element. Such access shall be provided either by means of a threaded cap for plug cocks or, for ball-type cocks, by a separable flanged or threaded two-section body.

3.5.2 Closure elements. The closure element shall be either a ported plug with spring and threaded cap or a ported ball with dual seat rings, at the option of the contractor. The closure elements shall be a commercial brass or bronze alloy.

3.5.2.1 Plug. Plugs shall be the tapered, inverted type accurately machined and finished to insure compliance with the leakage limitations of 3.4. Plug-type cocks shall include a spring designed and installed to exert a constant seating force against the bottom of the plug. Springs shall be phosphor bronze, austenitic stainless steel, or cadmium-plated steel spring wire. A threaded cap shall be provided on the bottom of the body to permit removal of the spring and plug. The cap shall be cadmium-plated steel or a commercial brass or bronze alloy and shall have paralled-faced external wrenching surfaces.

3.5.2.2 Ball. The ball shall rotate in replaceable seat rings fabricated of tetrafluoroethylene (TFE), trifluorochloroethylene (TFCE) or nitrile rubber. The stem shall be sealed with nonmetallic rings or packing to prevent external leakage exceeding the limits specified herein.

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3.5.3 Handle. Operating handles shall be the lever type and shall be malleable iron or steel. Handles shall be so installed that, when the plug or ball is in the closed position, the handle will be in line with the connected piping. The distance from the center of the operating nut to the end of the handle shall be not less than 3-1/4 inches for 1/2-inch cocks and 4-1/2 inches for 3/4 inch cocks. The handle shall include a lug to bear against the stops on the body. The handle ends on type II air cocks shall be upward inclined, at an angle between 20 to 30 degrees with the horizontal to provide adequate clearance from any standard air hose clamps on connected piping or fittings. The handle shall be labeled "OPEN, CLOSED" with arrow heads to indicate direction of motion to effect these actions.

3.6 Finish. All surfaces of the air cocks shall have the natural or machined finish normally produced by commercial manufacturing processes and techniques established as standard practice by the valve and fittings industry.

3.7 Marking. The plug cocks shall be marked in accordance with MSS SP-25.

3.7.1 Military marking. When specified (see 6.2), each air cock shall be labeled, stamped, or otherwise marked for identification in accordance with MIL-STD-130. Markings shall include information required by MIL-STD-130 or with portions thereof specified by the contracting officer (see 6.2).

3.8 Interchangeability. All air cocks of the same type, class, and size furnished under a specific contract or purchase order shall be identical to the extent required to insure interchangeability of renewable parts and to the extent required to permit interchangeable installation of air cocks once the required spacing in piping systems is established.

3.9 Type I. Type I air cocks shall be class 150 or class 300 as specified. Unless vented air cocks are specified (see 6.2), type I air cocks shall be the nonvented design. The rated air working pressure for vented air cocks shall be not less than 150 psig.

3.10 Type II. Type II air cocks shall be designed primarily for use on air brake systems of truck-tractors and similar automotive equipment. The air cocks shall be nonvented, shall be 1/2 inch nominal pipe size, and unless otherwise specified (see 6.2), shall be class 150. When specified (see 3.10.1 and 6.2), type II air cocks shall be equipped with a clamping stud, a 1/2 inch by 1/4 inch reducing bushing, or with both items.

3.10.1 Clamping stud. Clamping studs for type II air cocks shall consist of a heavy-duty close pipe nipple, a bushing, a plate washer, and a hexagon nut. The stud shall be designed to permit the air cock to be securely mounted on a chassis member or bulkhead on automotive equipment. The bushing shall protect the pipe threads within the mounting hole.

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3.11 Workmanship. The quality of workmanship shall be consistant with the level of quality established by the valve and fittings industry for air cocks and valves produced for commercial distribution. Castings shall be free from cracks, hot tears, blowholes, porosity, or other defects affecting structural soundness. Castings which must be plugged, impregnated, brazed, or burned-in to correct defects will not be acceptable. Inside and outside surfaces of castings shall be clean. Machined parts shall be free of cracks or other defects which will interfere with the proper functioning of the air cock. External surfaces shall be free of burrs, sharp edges and corners except where sharp edges or corners are required or where they are not detrimental to safety.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).
   (1) In-process examination.
   (2) End-item inspection.

c. Inspection of packaging (see 4.6).

4.3 First article inspection. First article inspection shall be performed on one air cock when a first article sample is required (see 3.2). This inspection shall include the examination of 4.5.1.1 and 4.5.1.2, and the tests specified in 4.5.2.1, 4.5.2.2 and 4.5.2.3. The sample air cock may be a standard production item from the contractor's current inventory provided the sample meets the requirements of this specification and is representative of the design, construction, and manufacturing techniques applicable to the remaining air cocks to be furnished under the contract or purchase order.

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4.4 Quality conformance inspection. Quality conformance inspection shall be as follows:

a. In-process inspection shall be performed on each air cock to be offered for acceptance by the Government and shall include the examination specified in 4.5.1.1 and the tests specified in 4.5.2.1 and 4.5.2.2.

b. End-item inspection shall be performed on samples selected in accordance with 4.4.1. This inspection shall include the examination of 4.5.1.2.

4.4.1 Sampling. A random sample of air cocks shall be selected from each lot offered to the Government in accordance with MIL-STD-105. The acceptable quality levels (AQL's) shall be 2.5 percent defective.

4.4.2 Lot. A lot shall consist of all air cocks of the same classification offered for delivery to the Government at one time. A sample unit shall consist of one completely assembled air cock.

4.5 Inspection procedure.

4.5.1 Examination.

4.5.1.1 In-process examination. Air cocks furnished under this specification shall be examined during production in accordance with the contractor's established quality control procedures which shall assure compliance with all requirements of this specification which can be checked visually and with those elements of the contractor's drawings or specifications normally verified during the inspection process. In-process examination shall apply to all areas of contract performance including the receipt and identification of material, fabrication of the components, finishing of parts, and final assembly of the air cocks. Quality control procedures shall be subject to surveillance by the contracting officer or designated representatives thereof. Failure of the contractor to demonstrate that adequate quality control was applied to lots submitted for acceptance by the Government may be cause for rejection of the lots.

4.5.1.2 End item inspection. Sample air cocks selected in accordance with 4.4.1 shall be examined to verify compliance with the nonoperational requirements of this specification. This examination shall verify that the units are complete and meet dimensional requirements and that all components and accessories are furnished as specified for the applicable classification of air cock. The end material, marking, and similar provisions are complied with. If the number of defective air cocks is equal to or greater than the rejection number applicable to the specified AQL, the lot represented by the sample units shall be rejected.

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4.5.2 Tests. Unless a commercial laboratory is used for inspection, the contractor shall furnish all measuring devices and test equipment and fixtures required to conduct the examination and tests specified herein. The quality control procedures specified in 4.5.1.1 shall provide for regular, calibration of instrumentation and test equipment in accordance with the contractor's established standards and procedures. Oil, grease, or other compounds shall not be applied to seating surfaces as a means of effecting a tighter seal.

4.5.2.1 Shell test. Each air cock shall be subjected to the pneumatic shell test pressure specified in 3.4. The pressure shall be applied with both ends blanked and the plug in the open position. The duration of the test shall be not less than 30 seconds. Any external leakage attributable to defects in workmanship or castings, or external leakage at the stem exceeding five cubic centimeters per hour, shall constitute failure of this test.

4.5.2.2 Seat tests. Each air cock shall be subjected to a pneumatic seat leakage test to verify compliance with 3.4. Tests shall be conducted using air as the test media. The test pressure specified in 3.4 shall be applied to one side of the plug. If no leakage occurs after the test pressure has been applied, the test may be discontinued. If measurable leakage occurs within the initial observation period, the test shall be continued for a length of time sufficient to permit an accurate determination of the leakage rate. Any leakage exceeding the maximum permissible rate specified in 3.4 shall be cause for rejection of the air cock unless corrective action to renew or recondition seating surfaces is taken and the air cock is satisfactorily retested. Soap suds applied to the discharge connection may be used to detect leakage. Formation of a one-inch soap bubble in less than 3 seconds shall constitute failure of the test.

4.5.2.3 Torque test. The first article sample shall be installed in a suitable test fixture and operated under rated working pressure to verify compliance with the closure torque requirements of 3.4. A torque wrench may be used for this test in lieu of the operating handle. The spring used on plug air cocks for the torque test shall have characteristics identical to springs used during seat leakage tests.

4.6 Inspection of packaging. The preservation, packing and marking of the air cocks shall be examined to determine compliance with the requirements of MIL-V-3, MIL-STD-129, and ASTM D3951.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

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5.1.1 Level A. The air cocks shall be preserved in accordance with the applicable level A preservation and packaging requirements of MIL-V-3.

5.1.2 Commercial. The air cocks shall be preserved in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. The air cocks shall be packed in accordance with the applicable level A requirements of MIL-V-3.

5.2.2 Level B. The air cocks shall be packed in accordance with the applicable level B requirements of MIL-V-3.

5.2.3 Commercial. The air cocks shall be packed in accordance with ASTM D3951.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D3951; additionally pack weight and cube shall be placed on the shipping container.

6. NOTES

6.1 Intended use. The air cocks covered by this specification are intended for use in plants and facilities for the control of air distribution in compressed air systems and to serve at construction sites for compressed air shutoff in lines to pneumatic tools. Type II cocks are also used in 125 psig (nominal) air brake systems on truck-tractors. In this application the cocks are installed in the service air line and emergency air line immediately ahead of the trailer brake connections and serve as manual shutoff devices to augment the shutoff action provided by the trailer protection valve in connection with two-way control valve. The type I and type II air cocks may also be used for gas, oil, and water service at temperatures to 150° F.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type required; size and class for type I; when type II shall be other than class 150 (see 1.2 and 3.10).

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c. When a first article is required for inspection and approval and the number of units required (see 3.2).

d. When military marking in accordance with MIL-STD-130 is required and information to be marked (see 3.7.1).

e. When type I air cocks shall be vented (see 3.9).

f. When a clamping stud, a reducing bushing, or both are required for type II air cocks (see 3.10).

g. Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the item should be examined and should be as specified in 4.3 or a sample selected from the first production run under the contract. The first article should consist of one or more units. The contracting officer should include specific instructions for examinations, test and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Custodians:
Army - ME
Air Force - 99

Review activities:
Air Force - 82
DLA - CS

User activity:
Army - CE

Preparing activity:
Army - ME

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-22546B	Cocks, Air, Cut-Out

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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